

Manual WordPress Setup on Ubuntu 20.04 Using Vagrant and Git Bash

Objective

This document explains how to manually install and configure WordPress on an Ubuntu 20.04 virtual machine using **Vagrant** and **Git Bash** on a Windows system.

Prerequisites

1. **Windows PC**
2. **Git Bash** – <https://git-scm.com/downloads>
3. **Vagrant** – <https://www.vagrantup.com/downloads>
4. **VirtualBox** – <https://www.virtualbox.org/wiki/Downloads>

Verify installation:

```
git --version
```

```
vagrant --version
```

Step 1: Initialize Vagrant VM

- 1) Create a project folder and navigate into it:

```
mkdir /f/vagrant-vm/wordpress_ubuntu
```

```
cd /f/vagrant-vm/wordpress_ubuntu
```

MINGW64/c:/Users/USER

```
USER@DESKTOP-V51CFFH MINGW64 ~ (main)
$ mkdir /f/vagrant-vms/wordpress_ubuntu
```

MINGW64/f/vagrant-vms/wordpress_ubuntu

```
USER@DESKTOP-V51CFFH MINGW64 ~ (main)
$ cd /f/vagrant-vms/wordpress_ubuntu/
```

2 Initialize Vagrant with Ubuntu 20.04:

vagrant init ubuntu/focal64

```
USER@DESKTOP-V51CFFH MINGW64 /f/vagrant-vm/wordpress_ubuntu
$ vagrant init ubuntu/focal64 --box-version 20240821.0.1
A `Vagrantfile` has been placed in this directory. You are now
ready to `vagrant up` your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
`vagrantup.com` for more information on using Vagrant.
```

3 vagrantfile settings to change

Got vagrant file cmd: `vim vagrantfile`

Vagrantfile Networking:

```
# Create a private network, which allows host-only access to the machine
# using a specific IP.
config.vm.network "private_network", ip: "192.168.50.11"

# Create a public network, which generally matched to bridged network.
# Bridged networks make the machine appear as another physical device on
# your network.
config.vm.network "public_network"
```

In your Vagrantfile, uncomment the following line to enable a private network with a fixed IP:

```
config.vm.network "private_network", ip: "192.168.50.11"
```

This allows you to access WordPress from your host machine using a consistent IP address.

If you need the VM to be accessible by other devices on your LAN, you can uncomment:

```
config.vm.network "public_network"
```

This makes your VM appear as a device on your local network.

Vagrant VM Provider Configuration: In your Vagrantfile, uncomment the VirtualBox provider block to customize VM settings:

```
#
| config.vm.provider "virtualbox" do |vb|
#   # Display the virtualBox GUI when booting the machine
#   vb.gui = true
#
#   # Customize the amount of memory on the VM:
vb.memory = "1024"
end
#
```

```
config.vm.provider "virtualbox" do |vb|
```

`config.vm.provider "virtualbox"` → Specifies that the following settings are **only for the VirtualBox provider**

```
vb.memory = "1024"
```

```
end
```

vb.memory = "1024" → Allocates **1024 MB of RAM** to the VM. This is the memory your Ubuntu VM will have to run WordPress.

Save and exit cmd : `esc+wq!`

3. Start the VM:

```
vagrant up
```

4. SSH into the VM:

```
vagrant ssh
```

5. Login to the root user

```
vagrant@ubuntu-focal:~$ sudo -i
```

```
root@ubuntu-focal:~$ sudo -i  
root@ubuntu-focal:~#
```

Step 2 . Install Dependencies

To install PHP and Apache, use following command:

```
sudo apt update  
sudo apt install apache2 \  
ghostscript \  
libapache2-mod-php \  
mysql-server \  
php \  
php-bcmath \  
php-curl \  
php-imagick \  
php-intl \  
php-json \  
php-mbstring \  
php-mysql \  
php-xml \  
php-zip
```

Step 3: Install WordPress

1)Instead of using Ubuntu's package, download the latest release directly from WordPress.org (recommended method).

```
sudo mkdir -p /srv/www
```

```
sudo chown www-data: /srv/www
```

```
curl https://wordpress.org/latest.tar.gz | sudo -u www-data tar zx -C /srv/www
```

Ownership is set to www-data. For multiple sites, use separate users and secure permissions (e.g., wp-config.php with chmod 400).

Use this command to check

- **ls -ld /srv/www/** → Shows details (permissions, ownership, size, date) of the /srv/www/ directory itself.
- **ls -l /srv/www/** → Lists all files and directories inside /srv/www/ with detailed information.
- **ls -l /srv/www/wordpress/** → Displays all WordPress core files and folders in the /srv/www/wordpress/ directory with details.

And Commands and Purpose

```
ls -ld /srv/www/
```

Shows detailed information **only about the /srv/www/ directory itself** (permissions, owner, group, size, and modification date).

Example:

text

```
drwxr-xr-x 3 www-data www-data 4096 Sep 9 19:14 /srv/www/
```

- drwxr-xr-x → permissions (read, write, execute flags)
- www-data www-data → owner and group
- 4096 → size in bytes of the directory
- Sep 9 19:14 → last modified date and time
- /srv/www/ → directory name

```
ls -l /srv/www/
```

Lists contents of /srv/www/ in **long format**, showing permissions, owners, groups, sizes, modification times, and names of all subfolders/files.

```
ls -l /srv/www/wordpress/
```

Lists **all WordPress core files and directories** within /srv/www/wordpress/. This includes:

- index.php → entry point
- license.txt → WordPress license
- readme.html → WordPress readme
- wp-admin → WordPress admin backend
- wp-content → themes, plugins, uploads
- wp-includes → core WordPress functions
- Various .php files like wp-config-sample.php, wp-login.php, etc.

Step 4. Configure Apache for WordPress

1 Create Apache site for WordPress.

```
vim /etc/apache2/sites-available/wordpress.conf
```

- Got to wordpress.conf using vim editor

Past following lines in to wordpress.conf

```
<VirtualHost *:80>

    DocumentRoot /srv/www/wordpress

    <Directory /srv/www/wordpress>

        Options FollowSymLinks

        AllowOverride Limit Options FileInfo

        DirectoryIndex index.php

        Require all granted

    </Directory>

    <Directory /srv/www/wordpress/wp-content>

        Options FollowSymLinks

        Require all granted

    </Directory>

</VirtualHost>
```

save and exit press esc and :wq

2 Enable the site with:

```
sudo a2ensite wordpress
```

```
root@ubuntu-focal:~# sudo a2ensite wordpress
Enabling site wordpress.
To activate the new configuration, you need to run:
    systemctl reload apache2
root@ubuntu-focal:~#
```

3 Enable URL rewriting with:

```
sudo a2enmod rewrite
```

```
root@ubuntu-focal:~# sudo a2enmod rewrite
Enabling module rewrite.
To activate the new configuration, you need to run:
    systemctl restart apache2
```

4 Disable the default “It Works” site with:

```
sudo a2dissite 000-default
```

```
root@ubuntu-focal:~# sudo a2dissite 000-default
Site 000-default disabled.
To activate the new configuration, you need to run:
    systemctl reload apache2
root@ubuntu-focal:~# |
```

5 Finally, reload apache2 to apply all these changes:

```
sudo service apache2 reload
```

```
root@ubuntu-focal:~# sudo service apache2 reload
root@ubuntu-focal:~#
```

Step 4. Configure database

To configure WordPress, we need to create MySQL database. Let's do it!

```
$ sudo mysql -u root
```

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 7

Server version: 5.7.20-0ubuntu0.16.04.1 (Ubuntu)

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> CREATE DATABASE wordpress;
```

Query OK, 1 row affected (0,00 sec)

```
mysql> CREATE USER wordpress@localhost IDENTIFIED BY '<your-password>';
```

Query OK, 1 row affected (0,00 sec)

```
mysql> GRANT SELECT,INSERT,UPDATE,DELETE,CREATE,DROP,ALTER
```

```
-> ON wordpress.*
```

```
-> TO wordpress@localhost;
```

Query OK, 1 row affected (0,00 sec)

```
mysql> FLUSH PRIVILEGES;
```

Query OK, 1 row affected (0,00 sec)

```
mysql> quit
```

Bye

Enable MySQL with `sudo service mysql start`

Step 5. Configure WordPress to connect to the database

6. Configure WordPress to connect to the database

Now, let's configure WordPress to use this database. First, copy the sample configuration file to wp-config.php:

```
sudo -u www-data cp /srv/www/wordpress/wp-config-sample.php /srv/www/wordpress/wp-config.php
```

Next, set the database credentials in the configuration file (*do not* replace database_name_here or username_here in the commands below. *Do* replace <your-password> with your database password.):

```
sudo -u www-data sed -i 's/database_name_here/wordpress/' /srv/www/wordpress/wp-config.php
```

```
sudo -u www-data sed -i 's/username_here/wordpress/' /srv/www/wordpress/wp-config.php
```

```
sudo -u www-data sed -i 's/password_here/<your-password>/' /srv/www/wordpress/wp-config.php
```

Finally, in a terminal session open the configuration file in nano:

```
sudo -u www-data nano /srv/www/wordpress/wp-config.php
```

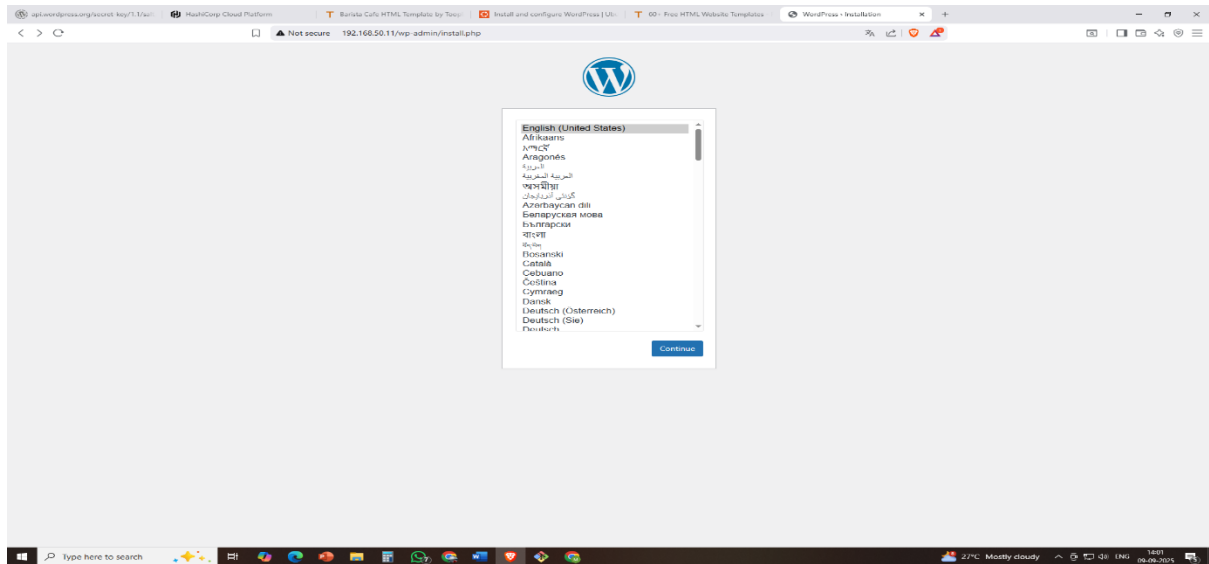
Find the following:

```
define( 'AUTH_KEY',      'put your unique phrase here' );
define( 'SECURE_AUTH_KEY', 'put your unique phrase here' );
define( 'LOGGED_IN_KEY',  'put your unique phrase here' );
define( 'NONCE_KEY',     'put your unique phrase here' );
define( 'AUTH_SALT',     'put your unique phrase here' );
define( 'SECURE_AUTH_SALT', 'put your unique phrase here' );
define( 'LOGGED_IN_SALT', 'put your unique phrase here' );
define( 'NONCE_SALT',    'put your unique phrase here' );
```

Delete those lines (ctrl+k will delete a line each time you press the sequence). Then replace with the content of <https://api.wordpress.org/secret-key/1.1/salt/>. (This address is a randomiser that returns completely random keys each time it is opened.) This step is important to ensure that your site is not vulnerable to “known secrets” attacks.

Save and close the configuration file by typing ctrl+x followed by y then enter

Step 6. Configure WordPress



Click enter continue

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Not secure 192.168.50.11/wp-admin/install.php?step=1



Welcome

Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title

Username
Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password [Hide](#)
Strong

Important: You will need this password to log in. Please store it in a secure location.

Your Email
Double-check your email address before continuing.

Search engine visibility ☐ Discourage search engines from indexing this site
It is up to search engines to honor this request.

[Install WordPress](#)

Fill the information and install the word press

api.wordpress.org/secret-key/1.1/x/ | HashCorp Cloud Platform | Barista Cafe HTML Template by Toop | Install and configure WordPress | Ub | 60+ Free HTML Website Templates | Log in - devops - WordPress

Not secure 192.168.50.11/wp-login.php



Username or Email Address

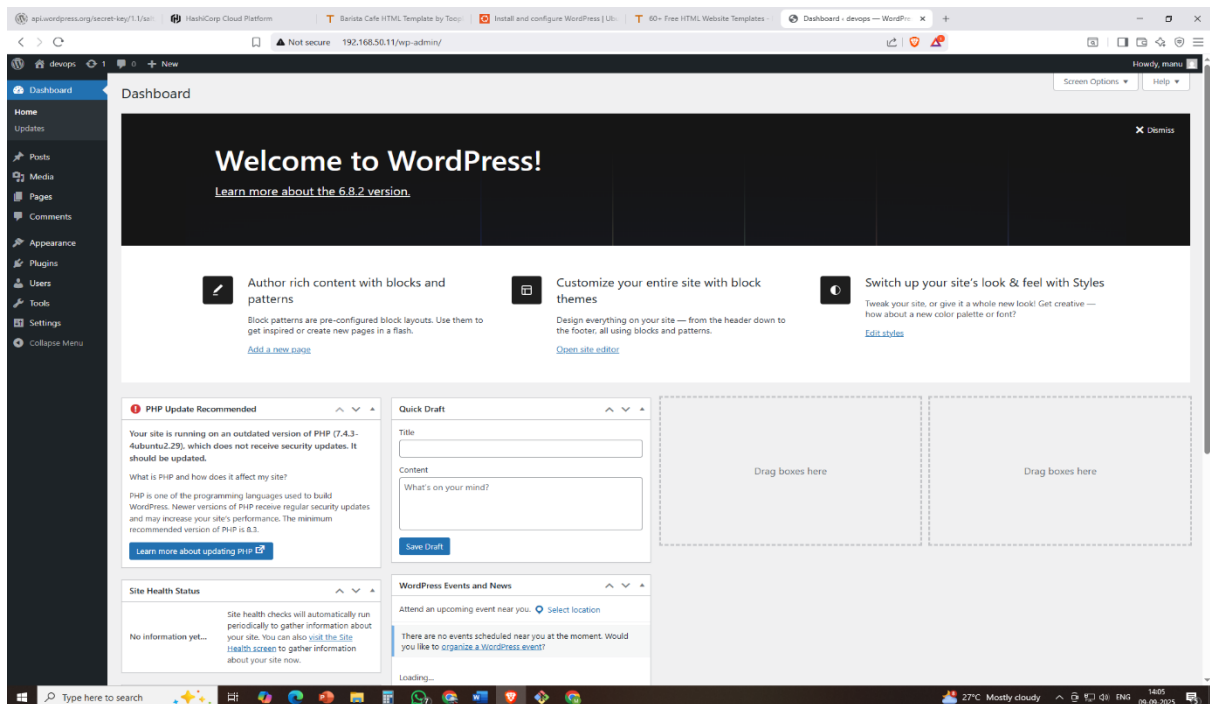
Password

 [Show](#)

☐ Remember Me [Log In](#)

Lost your password?
[Go to devops](#)

Login into wordpress



WordPress Installation – Command History

Update system packages

```
1 sudo apt update
```

Install Apache, MySQL, PHP, and required extensions

```
2 sudo apt install apache2 ghostscript libapache2-mod-php  
mysql-server \
```

```
php php-bcmath php-curl php-imagick php-intl php-json  
php-mbstring \
```

```
php-mysql php-xml php-zip
```

Create installation directory and set permissions

3 sudo mkdir -p /srv/www

4 sudo chown www-data: /srv/www

Download and extract the latest WordPress release

5 curl https://wordpress.org/latest.tar.gz | sudo -u www-data
tar zx -C /srv/www

Verify installation directory

8 ls -ld /srv/www/

9 ls -l /srv/www/

10 ls -l /srv/www/wordpress/

Configure Apache Virtual Host for WordPress

6 cat /etc/apache2/sites-available/wordpress.conf

11 vim /etc/apache2/sites-available/wordpress.conf

13 vim /etc/apache2/sites-available/wordpress.conf

Enable WordPress site and required Apache modules

12 sudo a2ensite wordpress

17 sudo a2enmod rewrite

18 sudo a2dissite 000-default

19 sudo service apache2 reload

Configure WordPress Database

20 sudo mysql -u root

Configure wp-config.php with DB credentials

21 sudo -u www-data cp /srv/www/wordpress/wp-config-sample.php /srv/www/wordpress/wp-config.php

22 sudo -u www-data sed -i 's/database_name_here/wordpress/' /srv/www/wordpress/wp-config.php

23 sudo -u www-data sed -i 's/username_here/wordpress/' /srv/www/wordpress/wp-config.php

24 sudo -u www-data sed -i 's/password_here/<your-password>/' /srv/www/wordpress/wp-config.php

26 sudo -u www-data sed -i 's/password_here/admin123/' /srv/www/wordpress/wp-config.php

27 vim /srv/www/wordpress/wp-config.php

28 sudo -u www-data vim /srv/www/wordpress/wp-config.php

Get server IP address for testing

29 ip addr show

```
root@ubuntu-focal:~# history
 1 sudo apt update
 2 sudo apt install apache2          ghostscript          libapache2-mod-php          mysql-server
    php                            php-bcmath            php-curl                    php-imagick                 php-intl
    php-json                      php-mbstring          php-mysql                   php-xml                     php-zip
 3 sudo mkdir -p /srv/www
 4 sudo chown www-data: /srv/www
 5 curl https://wordpress.org/latest.tar.gz | sudo -u www-data tar zx -C /srv/www
 6 cat /etc/apache2/sites-available/wordpress.conf
 7 clear
 8 ls -ld /srv/www/
 9 ls -l /srv/www/
10 ls -l /srv/www/wordpress/
11 vim /etc/apache2/sites-available/wordpress.conf
12 sudo a2ensite wordpress
13 vim /etc/apache2/sites-available/wordpress.conf
14 clear
15 Enable URL rewriting with:
16 clear
17 sudo a2enmod rewrite
18 sudo a2dissite 000-default
19 sudo service apache2 reload
20 sudo mysql -u root
21 sudo -u www-data cp /srv/www/wordpress/wp-config-sample.php /srv/www/wordpress/wp-config.php
22 sudo -u www-data sed -i 's/database_name_here/wordpress/' /srv/www/wordpress/wp-config.php
23 sudo -u www-data sed -i 's/username_here/wordpress/' /srv/www/wordpress/wp-config.php
24 sudo -u www-data sed -i 's/password_here/<your-password>/' /srv/www/wordpress/wp-config.php
25* sudo -u www-data sed -i 's/password_here/<your-password>/' /srv/www/wordpress/wp-config.php
26 sudo -u www-data sed -i 's/password_here/admin123/' /srv/www/wordpress/wp-config.php
27 vim /srv/www/wordpress/wp-config.php
28 sudo -u www-data vim /srv/www/wordpress/wp-config.php
29 ip addr show
30 history
root@ubuntu-focal:~#
```



This document records the exact command history used for setting up WordPress. It can serve as a reference guide for future projects and as part of my DevOps learning

Thank You for reading.